

# Strong-lens parameter inference: EM-only, GW-only, and EM+GW

## A cross-method comparison on a common mock

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## 1 System and simulation

All three cases fit the *same* mock lens (`shared/system_config.py`, seed 87651): an EPL+SHEAR deflector ( $\theta_E = 1.2''$ ,  $e_1 = 0$ ,  $e_2 = 0.1$ ,  $\gamma = 2$ , shear  $\gamma_1 = 0.1$ ,  $\gamma_2 = 0$ ) with a Sérsic source at  $(0.2, -0.05)''$  and a Sérsic lens light, imaged on a  $40 \times 40$  grid at  $0.1''/\text{px}$ , Gaussian PSF (FWHM  $0.067''$ ), background rms  $10^{-2}$ ,  $t_{\text{exp}} = 2200 \text{ s}$ . Peak SNR  $\approx 290$ . The GW sector adds four lensed images with observables = relative time delays (3) + effective luminosity distances (4); truth  $T_\star = 1.479 \times 10^7 \text{ s}$ ,  $d_L = 11214 \text{ Mpc}$ .

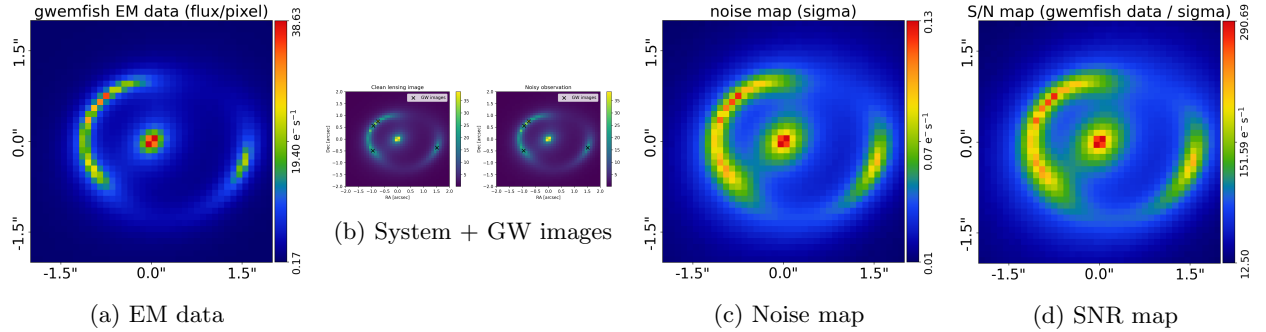


Figure 1: Simulated observation. (a) noisy EM image; (b) clean+noisy image with the 4 GW image positions ( $\times$ ); (c) per-pixel noise; (d) per-pixel SNR (peak  $\approx 290$ ).

**Measurement-error regimes (GW sector).** Two settings are compared for the GW cases: *large-error* ( $\sigma_{t_d} = 5\%$ ,  $\sigma_{d_L^{\text{eff}}} = 300\%$ ) and *precise* ( $\sigma_{t_d} = 0.1\%$ ,  $\sigma_{d_L^{\text{eff}}} = 5\%$ ). The simulated observables are identical; only the assumed uncertainties (posterior widths) change. EM-only is unaffected.

## 2 Case 1 — EM-only

**Free (11):**  $\theta_E, e_1, e_2, \gamma, \gamma_1, \gamma_2$  and source  $A, R_S, n, e_1, e_2$ . **Fixed to truth:** lens + source centres, shear origin, full lens-light profile, background noise. **Methods:** gwemfish deriv-approx (informed NUTS on a Taylor model) and fisher (Taylor–Gaussian at truth); PyAutoLens Nautilus; lenstronomy Nautilus. Priors: gwemfish/PAL use their native broad priors, lenstronomy uses truth  $\pm 10\sigma_{\text{Fisher}}$  boxes; at SNR $\sim 290$  the fit is data-dominated (verified by overlap).

**Result.** All four frameworks agree on every parameter to  $\lesssim 0.2\sigma$  in the means, with widths matching to a few percent (Table 1). The only outlier is source amplitude/radius ( $\sim 1\sigma$ ,  $\sim 0.2\%$ ): the irreducible Sérsic  $b_n$ -convention difference (PAL Ciotti–Bertin vs. HCL/lenstronomy). Common  $\sim 1$ – $2\sigma$  pulls ( $\theta_E - 1.2$ ,  $n + 2.1$ ) are the shared noise realization, not bias — the truth-centred Fisher posterior has identical widths.

Table 1: Case 1, representative parameters (mean  $\pm$  std; pull = (mean–truth)/std).

| param (truth)          | gwemfish deriv                    | gwemfish fisher                  | PyAutoLens                        | lenstronomy                       |
|------------------------|-----------------------------------|----------------------------------|-----------------------------------|-----------------------------------|
| $\theta_E$ (1.2)       | $1.1998 \pm 1.6\text{e-}4$ (−1.2) | $1.2000 \pm 1.6\text{e-}4$ (0.0) | $1.1998 \pm 1.6\text{e-}4$ (−1.2) | $1.1998 \pm 1.6\text{e-}4$ (−1.2) |
| $\gamma$ (2.0)         | $2.0005 \pm 5.2\text{e-}4$ (+0.9) | $2.0000 \pm 5.4\text{e-}4$ (0.0) | $2.0004 \pm 5.4\text{e-}4$ (+0.8) | $2.0004 \pm 5.4\text{e-}4$ (+0.8) |
| $e_2$ (0.1)            | $0.1000 \pm 3.8\text{e-}4$ (−0.1) | $0.1000 \pm 3.7\text{e-}4$ (0.0) | $0.1000 \pm 3.7\text{e-}4$ (−0.1) | $0.1000 \pm 3.8\text{e-}4$ (0.0)  |
| $n_{\text{src}}$ (1.5) | $1.5044 \pm 2.0\text{e-}3$ (+2.2) | $1.5000 \pm 2.0\text{e-}3$ (0.0) | $1.5039 \pm 2.0\text{e-}3$ (+2.0) | $1.5041 \pm 2.0\text{e-}3$ (+2.1) |

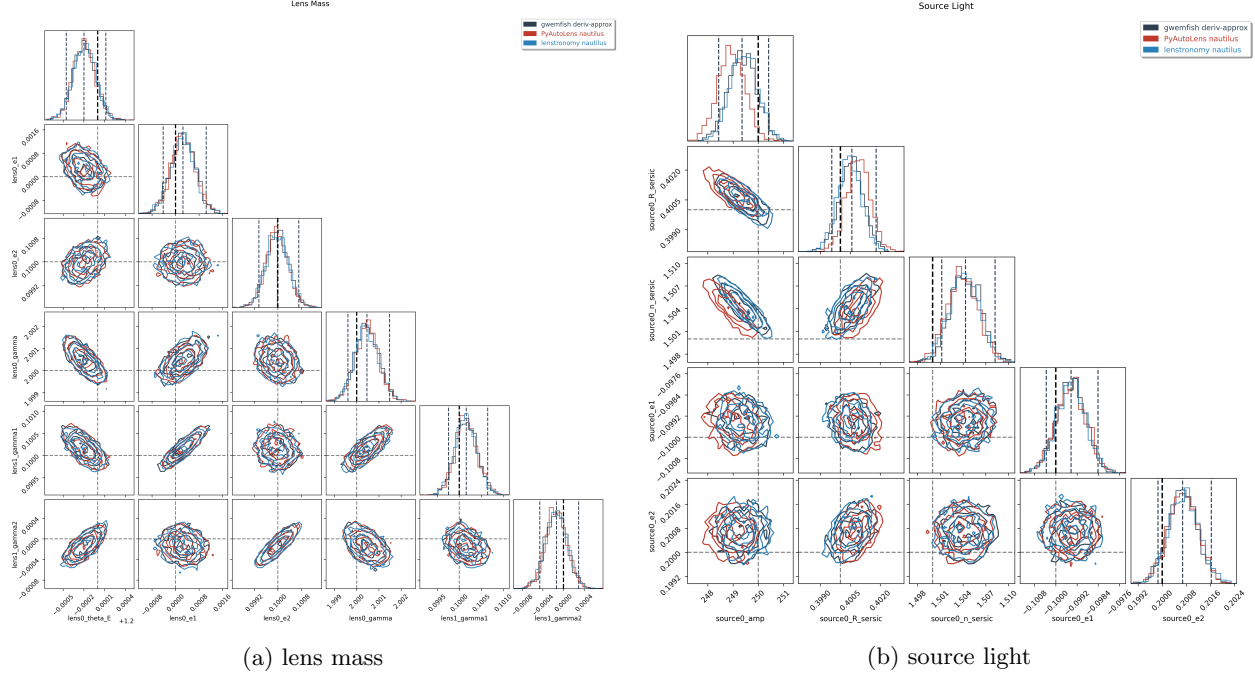


Figure 2: Case 1 EM-only: four-way comparison corners. Contours overlap to  $\lesssim 0.2\sigma$ .

### 3 Case 2 — GW-only

**Free (4):**  $e_2, \gamma, y_0^{\text{gw}}, y_1^{\text{gw}}$  (source position). **Fixed to truth:**  $\theta_E, e_1$ , lens centre, all shear,  $T_\star, d_L$ . Constraint = time delays ( $d_L^{\text{eff}}$  nearly uninformative in large-error). **Methods:** fisher-source, deriv-approx-source, nautilus-source (helens solver), and nautilus with a lenstronomy solver (same GW likelihood, different lens-equation solver — a cross-check). **Priors (Uniform):** NUTS boxes  $e_2 \in (-0.5, 0.5)$ ,  $\gamma \in (1.5, 2.5)$ ,  $y_0 \in (0.1, 0.3)$ ,  $y_1 \in (-0.13, 0.03)$ ; nautilus uses truth  $\pm 3\sigma_{\text{Fisher}}$  clipped to these (tight in precise, NUTS-dominated in large-error).

**Result.** Every method is consistent with truth (all  $|\text{pull}| \leq 0.7$ ). Precise data tighten the posteriors  $\sim 5\text{--}40\times$  (Table 2) and *resolve the  $y_1^{\text{gw}}$  bimodality* seen in large-error (Fig. 3). Fisher-source flips from a poor,  $\sim 10\times$ -too-wide approximation (near-singular FM, cond  $1.9 \times 10^9$ ) to an excellent one. The two solvers agree to  $\sim 10^{-8}$  nats except on a  $\sim 5\text{--}10\%$  caustic-boundary sliver each truncates differently ( $\leq 0.4\sigma$  mean shifts).

Table 2: Case 2 GW-only, posterior std (nautilus + lenstronomy solver); all means at truth.

| param             | large-error std | precise std | tighter         |
|-------------------|-----------------|-------------|-----------------|
| $e_2$             | 0.029           | 0.0032      | $\sim 9\times$  |
| $\gamma$          | 0.216           | 0.047       | $\sim 5\times$  |
| $y_0^{\text{gw}}$ | 0.0097          | 0.0011      | $\sim 9\times$  |
| $y_1^{\text{gw}}$ | 0.0186          | 0.0005      | $\sim 40\times$ |

### 4 Case 2f — GW-only with $T_\star, d_L$ free: a degeneracy and its resolution

**Motivation.** Case 2 fixes  $T_\star$  (time-delay distance) and  $d_L$  (luminosity distance) to truth. Freeing them — **free (6):**  $e_2, \gamma, y_0^{\text{gw}}, y_1^{\text{gw}}, T_\star, d_L$  — tests whether GW-only data can measure the distance/time-delay normalisation on their own, the quantity time-delay cosmography ultimately wants. **Methods:** fisher-source, deriv-approx-source (regularized informed NUTS, required once  $T_\star, d_L$  open a near-flat direction), nautilus-source (helens solver), nautilus + lenstronomy solver. **Priors:** Uniform truth  $\pm 50\%$  on  $T_\star, d_L$ ;

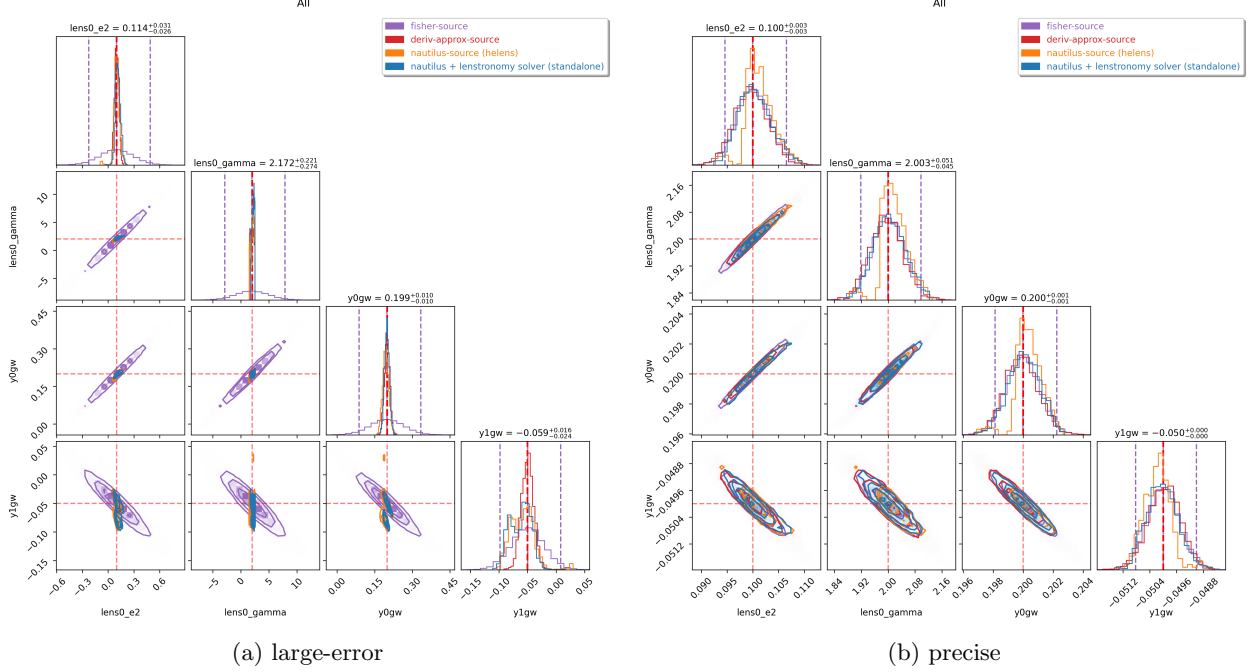


Figure 3: Case 2 GW-only comparison corners. Precise data collapse the widths and remove the secondary  $y_1^{\text{gw}}$  mode near  $-0.08$  visible at large error.

nautilus boxes from the fisher sigmas ( $\text{truth} \pm 3\sigma$ , clipped), as in Case 2.

**The degeneracy.** Freeing  $T_\star, d_L$  opens two near-flat Fisher directions:  $\text{corr}(d_L, \gamma) = -0.999$ ,  $\text{corr}(T_\star, y_1^{\text{gw}}) = +0.998$ ,  $\text{corr}(T_\star, d_L) = -0.97$ , with two unit-normalised Fisher eigenvalues at  $3.7 \times 10^{-11}$  and  $1.1 \times 10^{-9}$  (verified real rather than inversion noise: sigmas from the unit-normalised inversion match the fisher-meta sigmas to 4 decimal places). Physically,  $d_L^{\text{eff}} = d_L / \sqrt{\mu}$  ties  $d_L$  to the magnification, and time delays fix only the Fermat-potential scale up to the  $T_\star$  normalisation — GW observables alone under-determine which combination of  $T_\star, d_L, \gamma$  produced the data.

**Precise regime** ( $\sigma_{t_d} = 0.1\%$ ,  $\sigma_{d_L^{\text{eff}}} = 5\%$ ): **biased along the ridge.** The two exact-likelihood samplers (helens and lenstronomy solvers) agree with each other (means within  $\sim 0.5\sigma$ ) but both sit  $1.3\text{--}3.1\sigma$  down-ridge of truth (Fig. 4a) — a coherent shift along the degenerate direction, not a solver or sampler bug (all six pulls move together with the correlation signs above). deriv-approx-source, a Taylor surrogate centred at truth, stays truth-centred by construction and is the local-curvature answer, not the full-posterior one.

**scan\_opt regime** ( $\sigma_{t_d} = 1\%$ ,  $\sigma_{d_L^{\text{eff}}} = 0.5\%$ ): **resolved by re-allocating error, not by shrinking it.** Because  $d_L^{\text{eff}}$ , not  $t_d$ , carries the distance normalisation, trading back  $10\times$  on time-delay precision to buy  $10\times$  on  $d_L^{\text{eff}}$  collapses the ridge: every one of the 24 pulls across all four methods satisfies  $|p| \leq 0.62$  (Fig. 4b);  $d_L$  is measured to  $\sim 8\text{--}9\%$ ,  $T_\star$  to  $\sim 4\text{--}5\%$ ,  $\gamma$  to  $\sim 10\%$ , source position to  $\sim 2\%$ . This operating point was located in advance with a Fisher error-requirement scan; its analytic  $J^T C^{-1} J$  prediction matches the pipeline’s measured Hessian sigmas to five decimal places at the chosen budget, confirming the scan and the sampler compute the same object. fisher-source also becomes informative rather than a correlation-only source:  $\text{cond}(\text{FM})$  improves from  $\sim 10^{26}$ -class degeneracy (precise) to  $6.5 \times 10^{22}$  (scan\_opt).

**Caveats.** nautilus-helens has the fewest effective samples of the two exact solvers at scan\_opt ( $n_{\text{eff}} = 302$  vs  $3023$  for lenstronomy); its  $\sim 0.5\sigma$  residual pulls on  $\gamma, y_1^{\text{gw}}, T_\star$  are consistent with that sampling noise, not bias.  $\gamma$  remains largely prior-dominated at scan\_opt (GW-only constrains the  $\gamma\text{--}d_L$  combination, not  $\gamma$  alone). The  $T_\star\text{--}d_L$  ridge itself is not removed by scan\_opt, only bounded: all three sampled methods still trace the same anti-correlated band, now centred on truth. This degeneracy is exactly what Case 3 (EM+GW) breaks by an independent route — pinning the mass/source model with EM data rather than re-balancing GW error budgets.

Table 3: Case 2f, nautilus-helens: same total measurement effort, different  $\sigma_{t_d}/\sigma_{d_L^{\text{eff}}}$  split. Re-allocating error removes the bias.

| param (truth)                      | precise pull  | scan_opt pull  | scan_opt std         |
|------------------------------------|---------------|----------------|----------------------|
| $e_2$ (0.1)                        | +2.4 $\sigma$ | +0.03 $\sigma$ | 0.0103               |
| $\gamma$ (2.0)                     | +1.7 $\sigma$ | +0.50 $\sigma$ | 0.193                |
| $y_0^{\text{gw}}$ (0.2)            | +2.1 $\sigma$ | −0.45 $\sigma$ | 0.0047               |
| $y_1^{\text{gw}}$ (−0.05)          | −2.0 $\sigma$ | +0.62 $\sigma$ | 0.0076               |
| $T_\star$ ( $1.479 \times 10^7$ s) | −2.1 $\sigma$ | +0.49 $\sigma$ | $5.97 \times 10^5$ s |
| $d_L$ (11214 Mpc)                  | −3.1 $\sigma$ | −0.25 $\sigma$ | 1038 Mpc             |

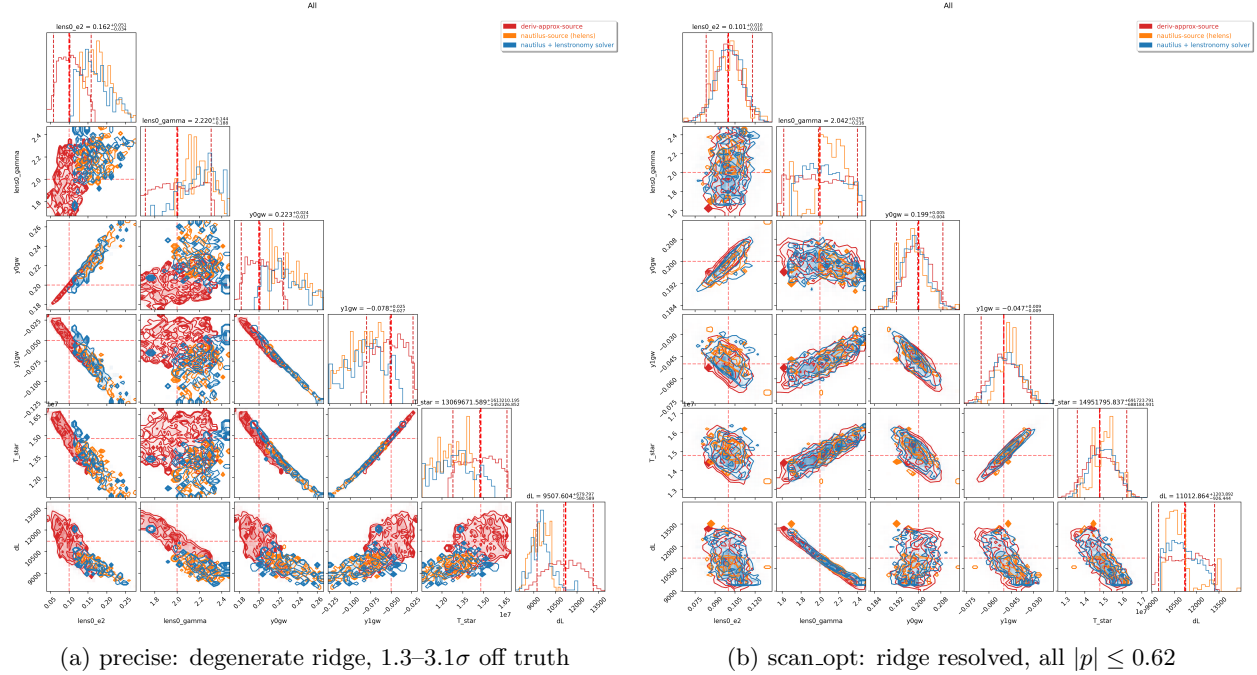


Figure 4: Case 2f, full six-parameter corner ( $e_2, \gamma, y_0^{\text{gw}}, y_1^{\text{gw}}, T_\star, d_L$ ), three methods overlaid (red deriv-approx-source, orange nautilus-helens, blue nautilus+lenstronomy), truth crosshairs dashed red. Re-allocating the same total measurement effort from time-delay to  $d_L^{\text{eff}}$  precision collapses the  $T_\star$ – $d_L$ – $\gamma$  degeneracy without adding data.

## 5 Case 3 — EM+GW

**Free (27):** full joint EM pixel model + GW sector ( $T_\star, d_L, y_0^{\text{gw}}, y_1^{\text{gw}}$ ). **Fixed to truth:** shear origin only. **Methods:** fisher-source, deriv-approx-source, nautilus-source (helens). **Priors:** lens-light centroid  $\mathcal{N}(0, 0.05)$ ,  $y_0/y_1 \sim \text{Uniform truth} \pm 0.05$ , rest free per parameter layout; nautilus boxes truth  $\pm 5\sigma_{\text{Fisher}}$ . (nautilus ties the GW source to the EM source centre, so its  $y_0/y_1$  column carries the EM astrometric constraint.)

**Result.** The EM likelihood already pins the mass and source-position parameters, so precise GW data change them negligibly; what they transform is the *GW-only sector* (Table 4):  $d_L$  goes from effectively unconstrained ( $\pm 11800$ ,  $\sim 70\%$ ) to  $\pm 288$  Mpc ( $\sim 2.6\%$ ), and  $T_\star$  tightens  $\sim 11\times$ . fisher and nautilus recover truth (nautilus  $|pull| \leq 1.2$ ; large pulls are std-amplification, not bias — absolute accuracy is sub-mas). *Caveat:* deriv-approx (banana surrogate) mixes poorly on the sharp precise 27-D posterior (worst  $\hat{R} \approx 1.9$ ); treat its contour as indicative — fisher and nautilus are the reliable methods here.



Table 4: Case 3 EM+GW, nautilus-source std: where precise GW data help.

| param             | large-error std   | precise std       | tighter         |
|-------------------|-------------------|-------------------|-----------------|
| $d_L$ (Mpc)       | 11800 (unconstr.) | 288               | $\sim 40\times$ |
| $T_\star$ (s)     | $4.9 \times 10^5$ | $4.6 \times 10^4$ | $\sim 11\times$ |
| $\gamma$          | 0.0030            | 0.0028            | $\sim 1\times$  |
| $y_1^{\text{gw}}$ | 0.00020           | 0.00019           | $\sim 1\times$  |

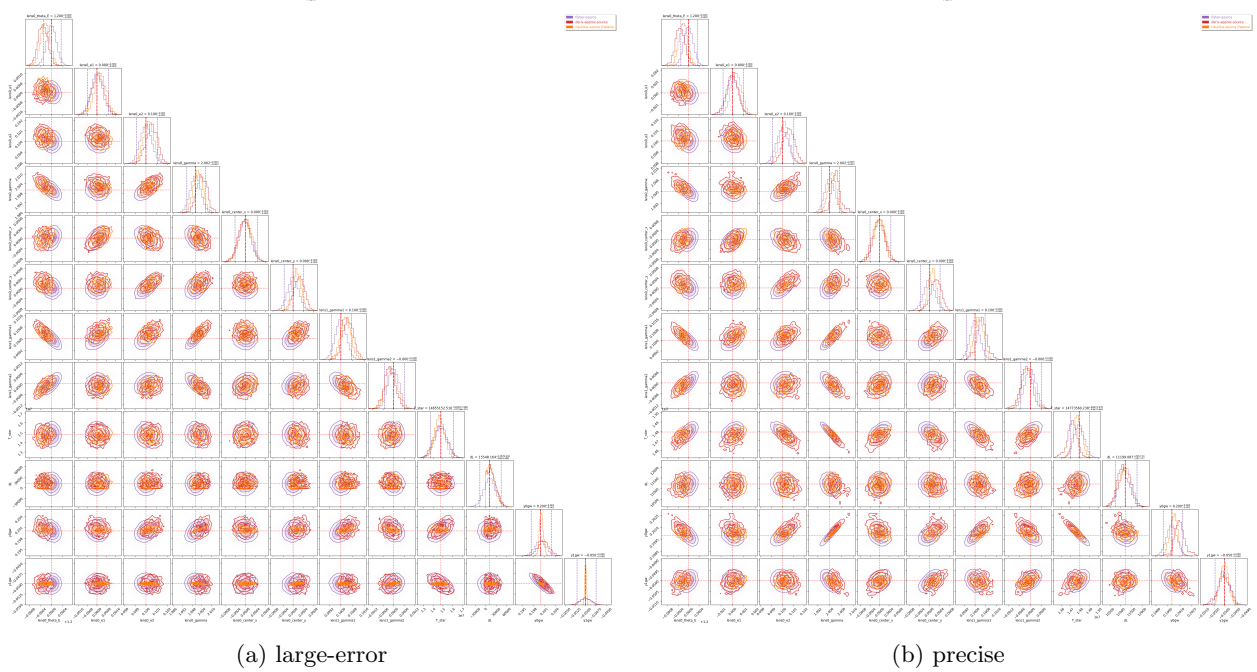


Figure 5: Case 3 EM+GW comparison corners (main parameters). Mass/source params are EM-dominated and stable; the visible tightening is in the GW sector ( $d_L, T_\star$ ).

## 6 Conclusion

On a common mock, independent codes and inference methods agree: (i) **EM-only** — gwemfish, PyAutoLens and lenstronomy recover truth to  $\lesssim 0.2\sigma$ , the only residual being the known Sérsic  $b_n$ -convention offset; (ii) **GW-only** — all methods are truth-consistent; precise timing shrinks the posterior  $\sim 5\text{--}40\times$  and resolves the  $y_1^{\text{gw}}$  bimodality, and Fisher becomes an accurate surrogate; (iii) **GW-only,  $T_\star, d_L$  free** — freeing the distance/time-delay normalisation opens a near-perfect  $T_\star\text{--}d_L\text{--}\gamma$  degeneracy (correlations  $\geq 0.97$ ) that biases every method coherently by  $1.3\text{--}3.1\sigma$  under a time-delay-optimised error budget; re-allocating the same total measurement effort toward  $d_L^{\text{eff}}$  precision (found in advance with a Fisher error-requirement scan) collapses all 24 pulls to  $|p| \leq 0.62$  and yields  $d_L$  to  $\sim 8\text{--}9\%$ ,  $T_\star$  to  $\sim 4\text{--}5\%$ , with zero extra data — the key lesson being that *how* error budget is spent matters as much as its total size, and a tight posterior is not necessarily a correct one; (iv) **EM+GW** — EM fixes the lens/source, so precise GW data act almost entirely on the GW-only sector, turning  $d_L$  from unconstrained into a  $\sim 2.6\%$  measurement — an independent route to breaking the same  $T_\star\text{--}d_L\text{--}\gamma$  degeneracy that (iii) resolves by GW error re-allocation alone. Cross-checks hold throughout (independent solvers agree to  $\sim 10^{-8}$  nats; vectorized vs. scalar likelihoods to  $\sim 10^{-11}$ ; analytic Fisher predictions match measured Hessian sigmas to 5 decimal places). The one method-level caveat is deriv-approx-source, whose banana-surrogate NUTS mixes poorly once a posterior becomes very sharp or opens a near-flat direction without regularization — fisher-source and nautilus-source remain reliable across all regimes tested.

This concludes the current phase of the comparison analysis: EM-only, GW-only (fixed and free  $T_\star, d_L$ , three error regimes), and EM+GW are all cross-validated across gwemfish, PyAutoLens and lenstronomy

on one common mock, with every run's samples, priors and configuration saved for exact reproduction (see `comparison-analysis/*/results*.md` and `handoff.md`). Open follow-ups (not required for this conclusion) are recorded in `comparison-analysis/handoff.md`: a mac-environment (herculens 0.3.x) rerun to confirm absolute numbers, and a Case 3 nautilus-source variant with a freed GW source position.